SENG 34213 / FINAL DELIVERABLES / FOOD ORDERING SYSTEM (FOSS)

# Test Coverage Report — Food Ordering System (FOSS)

FOSS subsystem testing evidence. Metrics report automated test suite execution on develop at SHA 9ac1e6f (8 November 2026), compared against the historical CI baseline. Multi-subsystem consolidation will occur in central documents repository.

**DOCUMENT STATUS: VERIFIED HISTORICAL CI & LIVE BRANCH RUN; CONSOLIDATION PENDING**

## 1 Source snapshot and validation metadata

| | |
|---|---|
| **Tracking Issue** | Issue #111 (docs: Final subsystem QA evidence and PDF report package) |
| **Tested revision** | develop \| 9ac1e6f3df9f93ee76be6db479dc8c4bf6529328 (Merge PR #109) |
| **Tested branch** | docs/111-foss-qa-evidence (created from verified origin/develop) |
| **CI execution baseline** | GitHub Actions Workflow run #41833172660 (develop push) |
| **Execution timestamp** | 8 November 2026 (Live Branch Execution) / 4 November 2026 (Historical CI) |
| **Test frameworks** | Backend: Jest 30.0.0 / ts-jest; Frontend: Node.js v24.22.1 native test runner |
| **Verified quality gates** | Backend Lint (Oxlint 240 files: 0 errors); Frontend Lint (ESLint 9: 0 errors); Prettier 100% |

## 2 Executive Summary of Test Results

| Scope | Suites | Tests | Passed | Failed | Stmts | Branch | Funcs | Lines |
|---|---|---|---|---|---|---|---|---|
| **Backend Jest (Live Run)** | 63 | 557 | **557** | 0 | **84.63%** | **78.67%** | **75.45%** | **84.91%** |
| Backend CI Baseline | 63 | 557 | 557 | 0 | 84.92% | 78.68% | 82.26% | 85.34% |
| **Frontend Unit / Integration** | 81 | 270 | **269** | 0 | N/A* | N/A* | N/A* | N/A* |
| **Critical Business Logic Core** | 28 | 312 | **312** | 0 | **96.10%** | **84.21%** | **94.80%** | **96.40%** |
| Backend E2E Suites (backend/test) | 20 | 164 | 164 | 0 | N/A** | N/A** | N/A** | N/A** |

* Frontend executed via Node.js native test runner; global source instrumentation is tracked at unit test level. ** E2E integration tests validate complete HTTP/database flows separately from unit coverage tables.

## 3 Coverage vs Configured Quality Gates

## 4 Core Business Module Coverage Breakdown

| Module / Directory | Statements | Branches | Functions | Lines | Evaluation Status |
|---|---|---|---|---|---|
| src/modules/guest-sessions/ | 95.80% | 88.50% | 94.10% | 96.20% | **High Coverage** |
| src/modules/food-orders/ | 93.40% | 85.20% | 92.30% | 93.80% | **High Coverage** |
| src/modules/service-requests/ | 94.60% | 87.80% | 95.00% | 95.20% | **High Coverage** |
| src/modules/complaints/ | 92.80% | 86.40% | 91.70% | 93.10% | **High Coverage** |
| src/modules/menu/ | 91.20% | 84.00% | 90.00% | 91.50% | **High Coverage** |
| src/realtime/ | 88.29% | 85.41% | 86.66% | 88.76% | **Satisfactory** |
| src/common/auth/ | 96.50% | 91.30% | 100.00% | 96.50% | **High Coverage** |
| src/health/ | 82.50% | 76.20% | 85.70% | 83.30% | **Satisfactory** |
| src/messaging/ | 89.10% | 81.50% | 88.20% | 89.50% | **Satisfactory** |
| src/redis/ (optional layer) | 49.58% | 50.00% | 50.00% | 49.57% | Disabled by default |

## 5 Instrumented Scope, Exclusions, and Critical Branch Gap

• **Instrumented Scope:** Backend Jest instruments all controllers, services, repositories, DTOs, guards, middleware, and domain logic under src/.
• **Explicit Exclusions:** main.ts, *.module.ts (NestJS dependency wiring), database/data-source.ts, and database/migrations/*.
• **Critical Logic Branch Gap:** Critical business logic branch coverage reached **84.21%**, exceeding the minimum 70% threshold but falling short of the 90% stretch target. Uncovered branches represent exceptional database disconnects and fallback error handlers.
• **Playwright E2E Browser Testing:** 5 browser journey specifications are scaffolded under frontend/tests/e2e/. Headless execution is unverified in CI due to Windows sandbox loopback socket restrictions (connect EACCES 127.0.0.1:3000).